

Enterprise Network Implementation Using Cisco Packet Tracer

A Dissertation Submitted To

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In Partial Fulfillment Of The Requirement for
The Award Of The Degree Of

COMPUTER NETWORKING

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01 | Enterprise Network Infrastructure Documentation

Project identity, scope and document control

ENTERPRISE HQ + BRANCH NETWORK

Cisco Packet Tracer implementation documented as a reproducible engineering project covering routing, switching, VLAN segmentation, OSPF, DHCP, ACLs, SSH, EtherChannel/LACP, Port Security, DHCP Snooping, Dynamic ARP Inspection, Wi-Fi, IoT, NAT/PAT and a self-contained Internet simulation.

This edition is intentionally different from the earlier sparse 40-page layout. The content is reflowed so that every page carries a substantial amount of useful engineering information. Section boundaries continue naturally from page to page instead of leaving large empty areas.

The project has two principal sites. The headquarters uses a multilayer switch as its Layer-3 aggregation point. The branch uses two access switches with an LACP EtherChannel and a router-on-a-stick design. The two sites meet through R-ISP, and R-INTERNET simulates the public Internet through a loopback at 8.8.8.8.

Document field	Value
Project type	Enterprise multi-site network simulation
Platform	Cisco Packet Tracer lab implementation
Core routing model	Layer-3 switching + routed WAN/edge routers
WAN/Internet model	OSPF internal routing + NAT/PAT + simulated 8.8.8.8 loopback
Security model	Segmentation, Guest isolation, management hardening and edge controls
Final status	Operational; major planned services validated

Reading this document

The first half explains the architecture and configuration, the middle sections document wireless, IoT and troubleshooting, and the final sections provide operational verification, recovery guidance, and a project acceptance summary.

The documentation uses the verified terminology and addressing from the build record. Commands shown later are limited to syntax that was accepted in the Packet Tracer image used for this project.

Document control note

Use this page as the project baseline. Record changes against the saved Packet Tracer file and keep the final acceptance outputs together with the document.

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02 | Executive Summary

What the project delivers at a glance

The project implements a realistic enterprise campus plus branch network with deliberate separation of responsibilities. At HQ, SW-CORE owns the user-facing SVI gateways for VLANs 10, 20, 30, 40, 50, 60, 70, 80, 90 and 99. At the branch, R-BR owns subinterfaces for VLANs 10, 40, 60, 80, 90 and 99 and serves as the branch DHCP server.

OSPF process 10 and area 0 provide dynamic exchange across the routed infrastructure. R-ISP acts as the edge translation router. Private sources from the 10.10.0.0/16 and 10.20.0.0/16 ranges are translated to the outside interface on R-ISP. R-INTERNET presents a stable 8.8.8.8/32 test endpoint.

Service	Final state
HQ VLAN segmentation	Operational
Branch VLAN segmentation	Operational
OSPF routing	FULL adjacency / learned routes
Branch DHCP	Verified leases across multiple VLANs
Guest ACL	Internal 10.10.x.x access blocked as designed
HQ Wi-Fi	10.10.90.x clients validated
Branch Wi-Fi	10.20.90.x clients validated
HQ IoT	10.10.80.x clients validated
NAT/PAT	Dynamic translations observed
Internet test	8.8.8.8 reachable from validated clients

The final validation record is especially important because the build encountered real troubleshooting events: branch DHCP requests initially failed, SW-BR2 Fa0/4 entered err-disabled because of DHCP Snooping rate-limit behavior, AP-BR was later found in VLAN 1 rather than VLAN 90, and HQ Internet traffic required explicit default routes and expanded NAT source coverage.

Project outcome

The network is not merely configured on paper. The final state was validated from endpoints through gateways, routed hops, NAT translations and the simulated Internet destination.

Executive interpretation

The architecture is intentionally end-to-end: segmentation, routing, security, wireless and NAT are connected services. A change in one layer should be validated before the next layer is modified.

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03 | Project Objectives and Requirements

Functional targets, engineering constraints and success criteria

The project objectives were defined as a coherent enterprise design rather than as isolated configuration exercises. The goal was to demonstrate a complete path from endpoint access through switching, routing, security enforcement and Internet egress.

Requirement group	Target
Segmentation	HQ VLANs 10/20/30/40/50/60/70/80/90/99 and branch VLANs 10/40/60/80/90/99
Routing	HQ inter-VLAN routing on SW-CORE; branch router-on-a-stick on R-BR
Dynamic routing	OSPF process 10, area 0
Address services	Branch DHCP pools for all six branch client VLANs
Security	Guest ACL, SSHv2, Port Security, PortFast/BPDU Guard, DHCP Snooping and DAI
Wireless	Separate HQ Wi-Fi, branch Wi-Fi and IoT wireless services
Internet	NAT/PAT through R-ISP with 8.8.8.8 simulation

A second objective was repeatability. During the build, several common Cisco command forms were rejected by the specific Packet Tracer image. Instead of treating those failures as operator mistakes indefinitely, the final procedure was normalized around the forms that actually worked, including ``show running-config``, ``show interfaces status``, ``show vlan brief``, ``show interfaces trunk``, and ``show ip dhcp snooping``.

A third objective was dependency-aware implementation. Trunks and EtherChannel were stabilized before DHCP and wireless validation. OSPF adjacency and routing tables were checked before Internet testing. Wireless clients were judged only after the AP profile and switch access VLAN were confirmed. This build order reduced the risk of changing several layers simultaneously.

Acceptance principle

A service is considered complete only when the configuration state and the observed behavior agree. A DHCP pool without a lease, or NAT configuration without a translation, is treated as incomplete until traffic proves the service is functioning.

Requirements traceability

Each stated objective maps to at least one observable configuration or test. This makes the project suitable for portfolio review because every feature has a verification method.

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04 | Final Topology

Physical and logical view of the finished Packet Tracer network

Topology evidence

The final physical view shows R-HQ, R-ISP, R-BR and R-INTERNET in the routed path, with SW-CORE at HQ, SW-BR1/SW-BR2 at the branch, AP-HQ for HQ user Wi-Fi, AP-BR for branch Wi-Fi, and AP-IOT for the dedicated IoT network.

The final topology contains R-HQ, R-ISP, R-BR and R-INTERNET in the routed path. SW-CORE provides the HQ Layer-3 boundary. SW-BR1 and SW-BR2 provide the branch access layer. AP-HQ carries HQ user Wi-Fi, AP-BR carries branch Wi-Fi, and AP-IOT carries the dedicated HQ IoT segment.

Traffic class	Forwarding sequence
HQ wired / Wi-Fi	Endpoint -> SW-CORE -> R-HQ -> R-ISP -> R-INTERNET
Branch wired / Wi-Fi	Endpoint -> SW-BR1/SW-BR2 -> R-BR -> R-ISP -> R-INTERNET
HQ IoT	IoT endpoint -> AP-IOT -> SW-CORE Fa0/9 (VLAN 80) -> routed backbone
Branch inter-switch	SW-BR1 <-> SW-BR2 using LACP Port-channel4

The topology is intentionally layered. Access devices attach to access ports; inter-switch and router links use trunks or routed point-to-point links as appropriate. This role separation is what later made DHCP and Internet troubleshooting manageable.

Topology reading guide

Read the diagram from endpoint toward the Internet. At every hop, identify the VLAN or routed subnet, then identify the next-hop device and the expected control-plane behavior.

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05 | Architecture Design

Layered enterprise model and failure domains

The architecture uses clear responsibilities. SW-CORE is the aggregation and routing boundary for HQ. Its SVIs provide the first-hop gateway for each HQ VLAN, and its routed uplink toward R-HQ becomes the northbound path to the WAN edge.

The branch access layer is split between SW-BR1 and SW-BR2. Their inter-switch links form Port-channel4 using LACP, which provides logical bundling of the parallel links. R-BR terminates the branch VLANs with 802.1Q subinterfaces on G0/0. This creates a simple and deterministic branch routing model.

R-HQ, R-ISP and R-BR form the internal routed backbone. R-ISP additionally connects to R-INTERNET through the 10.10.255.0/30 segment. The Internet simulator uses Loopback0 = 8.8.8.8/32, allowing the entire lab to remain self-contained.

Layer	Primary components	Main responsibility
Access	SW-BR1, SW-BR2, AP-HQ/AP-BR/AP-IOT	Endpoint access, VLAN membership, edge security
Aggregation / Core	SW-CORE	SVIs, inter-VLAN routing, OSPF, HQ default path
WAN / Edge	R-HQ, R-ISP, R-BR	Routed transit, default paths, DHCP at branch
Internet simulation	R-INTERNET	Stable 8.8.8.8/32 ICMP endpoint

Failure-domain concept

A client issue can be isolated to an edge port or AP VLAN, a route issue can be isolated to a routed hop, and an Internet issue can be traced separately through the NAT boundary. The architecture was designed to make those distinctions visible.

Design rationale

The architecture separates Layer-2 transport, Layer-3 gateway duties, dynamic routing and Internet translation. This creates smaller fault domains and clearer ownership of each function.

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06 | IP Addressing Plan

Enterprise, branch and transit addressing

HQ segment	VLAN	Subnet	Gateway
ADMIN	10	10.10.10.0/24	10.10.10.1
HR	20	10.10.20.0/24	10.10.20.1
FINANCE	30	10.10.30.0/24	10.10.30.1
IT	40	10.10.40.0/24	10.10.40.1
SALES	50	10.10.50.0/24	10.10.50.1
GUEST	60	10.10.60.0/24	10.10.60.1
SERVER-DMZ	70	10.10.70.0/24	10.10.70.1
IoT	80	10.10.80.0/24	10.10.80.1
WIFI	90	10.10.90.0/24	10.10.90.1
MANAGEMENT	99	10.10.99.0/24	10.10.99.1

Branch segment	VLAN	Subnet	Gateway
USERS	10	10.20.10.0/24	10.20.10.1
IT	40	10.20.40.0/24	10.20.40.1
GUEST	60	10.20.60.0/24	10.20.60.1
IoT	80	10.20.80.0/24	10.20.80.1
WIFI	90	10.20.90.0/24	10.20.90.1
MGMT	99	10.20.99.0/24	10.20.99.1

Routed link	Local / peer
R-BR G0/1 <-> R-ISP G0/1	10.10.252.1/30 <-> 10.10.252.2/30
R-HQ G0/1 <-> R-ISP G0/0	10.10.253.1/30 <-> 10.10.253.2/30
SW-CORE <-> R-HQ G0/0	10.10.254.2/30 <-> 10.10.254.1/30
R-ISP G0/2 <-> R-INTERNET G0/0	10.10.255.1/30 <-> 10.10.255.2/30
R-INTERNET Loopback0	8.8.8.8/32

The addressing plan uses /24 networks for user-facing VLANs so that each broadcast domain is easy to recognize and troubleshoot. Point-to-point /30 networks are used for routed links because the project requires only two usable endpoints on each transit segment.

Addressing discipline

Use the first host address as the gateway consistently. Reserve infrastructure addresses, keep point-to-point links on /30s, and use the subnet itself as the primary clue during troubleshooting.

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07 | VLAN Design and Segmentation

Business roles, trust boundaries and policy intent

VLAN	Name	Primary purpose	Security intent
10	ADMIN / USERS	General users and admin endpoints	Standard user access
20	HR	Human Resources systems	Limit lateral exposure
30	FINANCE	Finance workstations	Sensitive data isolation
40	IT	Technical/admin endpoints	Privileged operational segment
50	SALES	Sales endpoints	Departmental separation
60	GUEST	Untrusted guest devices	Restricted by ACL
70	SERVER-DMZ	Server services	Server isolation / DMZ-style boundary
80	IoT	IoT sensors/devices	Dedicated device segment
90	WIFI	Wireless clients	Dedicated wireless client segment
99	MANAGEMENT	Network management	Administrative control plane

Segmentation is the central policy structure of the project. An endpoint address immediately exposes its logical security domain, allowing ACL rules, DHCP scopes and troubleshooting expectations to be mapped to a subnet.

Guest is intentionally untrusted. Management is separated from user traffic. IoT has its own broadcast domain instead of sharing the user Wi-Fi subnet. This segmentation makes later access-control improvements straightforward because the source and destination networks are explicit.

Why VLANs matter here

The design is not using VLANs merely to reduce broadcast traffic. Each VLAN represents a role, trust boundary or service category. Routing between VLANs is therefore a controlled policy decision, not an accidental consequence of a flat switch fabric.

Segmentation discipline

A VLAN is treated as a security and operational boundary. Any future ACL, DHCP, monitoring or QoS policy can target the VLAN subnet without redesigning the core addressing plan.

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08 | SW-CORE Configuration

Layer-3 switching, SVIs and HQ routing boundary

SW-CORE is the central Layer-3 switch for the HQ environment. `ip routing` is enabled so the switch can route between SVIs locally and forward remote destinations toward R-HQ. Each SVI uses the .1 address of its VLAN subnet, creating a consistent first-hop convention.

```
ip routing
interface Vlan10
ip address 10.10.10.1 255.255.255.0
interface Vlan20
ip address 10.10.20.1 255.255.255.0
...
interface Vlan99
ip address 10.10.99.1 255.255.255.0
```

All ten required HQ SVIs were verified as operational. The routing table showed connected entries for the 10.10.x.0/24 networks, while OSPF supplied the routed backbone and branch information. This is the expected division of labor: directly connected routes for local VLANs and dynamic routes for remote networks.

SW-CORE also participates in OSPF area 0 over the routed 10.10.254.0/30 link. Later, a static default route was added toward R-HQ to provide an explicit northbound Internet exit.

```
router ospf 10
router-id 4.4.4.4
network 10.10.254.0 0.0.0.3 area 0
network 10.10.10.0 0.0.0.255 area 0
...
network 10.10.99.0 0.0.0.255 area 0
ip route 0.0.0.0 0.0.0.0 10.10.254.1
```

Verification focus

Use `show ip interface brief` to verify SVI state, `show ip route` to verify connected and OSPF routes, and `show ip ospf neighbor` to confirm the core-to-R-HQ relationship.

Core verification

Confirm that SVIs are up, routing is enabled, OSPF neighbors are established and the static default exists. If one of these is missing, downstream client testing will produce misleading symptoms.

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09 | Branch Switching Design

SW-BR1, SW-BR2 and validated access roles

Device	Port	VLAN	Role / endpoint context
SW-BR1	Fa0/1	10	BR user
SW-BR1	Fa0/2	10	BR user
SW-BR1	Fa0/3	40	BR IT
SW-BR2	Fa0/1	10	BR user
SW-BR2	Fa0/2	10	BR user
SW-BR2	Fa0/3	60	BR guest
SW-BR2	Fa0/4	90	BR Wi-Fi client context
SW-BR2	Fa0/10	90	AP-BR uplink
SW-CORE	Fa0/9	80	AP-IOT uplink

The access assignments reflect the final validated topology rather than an abstract design. SW-BR2 Fa0/10 is particularly important because it became the AP-BR uplink during troubleshooting. When that port remained in VLAN 1, the branch wireless DHCP requests could not reach the VLAN 90 gateway.

Connected interfaces were preserved during hardening. The build did not shut a port merely because a device was temporarily quiet. A disconnected state had to be established before an unused port was considered for shutdown. This protects required APs, routers and trunks from accidental administrative changes.

Both branch switches carry VLANs 10, 40, 60, 80, 90 and 99 across the inter-switch trunk. This common VLAN set is what lets branch endpoints reach the appropriate router subinterface regardless of whether they attach to SW-BR1 or SW-BR2.

Switching verification sequence

Start with ``show vlan brief``, then ``show interfaces trunk``, then ``show etherchannel summary``. When a client has the wrong IP or no DHCP lease, confirm the physical access VLAN before inspecting higher-layer services.

Access-layer verification

Confirm the endpoint port VLAN before checking the router. A switchport in the wrong VLAN can produce a DHCP failure even when the router pool and subinterface are perfectly configured.

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10 | EtherChannel / LACP

Port-channel4 design, benefits and verification

SW-BR1 and SW-BR2 are joined with two physical links that operate as one logical EtherChannel. The final implementation uses LACP in active mode on Fa0/23 and Fa0/24, with the bundle represented by Port-channel4.

```
interface FastEthernet0/23
switchport trunk allowed vlan 10,40,60,80,90,99
switchport mode trunk
channel-group 4 mode active
interface FastEthernet0/24
switchport trunk allowed vlan 10,40,60,80,90,99
switchport mode trunk
channel-group 4 mode active
interface Port-channel4
switchport trunk allowed vlan 10,40,60,80,90,99
switchport mode trunk
```

The validated state showed Group 4 with LACP and Fa0/23(P), Fa0/24(P) as bundled members. The `(P)` state is significant because it indicates the physical interfaces are participating in the logical port channel rather than acting as unrelated parallel links.

Benefit	Engineering effect
Loop prevention	STP sees the logical bundle instead of two independent parallel Layer-2 paths
Availability	A member link can fail while the Port-channel remains available
Bandwidth	Both physical members can contribute to the logical link
Operational clarity	One logical interface is used for trunk verification and VLAN forwarding

LACP was therefore a design decision, not a decorative feature. The branch topology depends on the port-channel to keep the VLAN set stable between the two access switches.

Bundle verification

Both physical members must be in the same EtherChannel and use compatible trunk settings. Verify the logical Port-channel, not only the individual links, before changing STP or VLAN configuration.

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11 | Spanning Tree and Loop Prevention

PVST, PortFast and BPDU Guard

The switches use PVST with system-ID extension. During VLAN 90 validation, SW-BR1 was observed as the root for the VLAN while SW-BR2 used Port-channel4 as its root port. That state demonstrates normal Layer-2 convergence across the access layer.

```
interface FastEthernet0/1
spanning-tree portfast
spanning-tree bpduguard enable
```

PortFast reduces the amount of time a host-facing interface spends waiting for spanning-tree convergence. BPDU Guard protects that same edge port by reacting when an unexpected bridge protocol data unit appears. Together they reduce the risk of a host port being used to form a Layer-2 loop or introduce a switch where a simple endpoint was expected.

Port role	Recommended treatment in this project
Host-facing access port	PortFast + BPDU Guard + access VLAN
Inter-switch trunk / Port-channel	Do not treat as a simple edge port
Router uplink	Routed or trunk role; preserve normal control-plane behavior
AP access port	Access VLAN selected for the SSID/service segment

An important lesson from the project is role-based configuration. A command that is correct on a PC-facing edge interface can be harmful when copied to a trunk or router uplink. The final configuration kept edge hardening aligned to the physical role of the port.

Loop-prevention rule

Apply edge protections only to edge ports. Preserve normal STP behavior on trunks and routed links. When an edge port enters err-disabled, inspect the specific protection feature that triggered it.

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12 | OSPF Design

Dynamic routing across the enterprise

OSPF process 10 and area 0 provide the internal routing control plane. The project uses compact /30 routed segments so that each point-to-point adjacency is easy to identify and verify.

Device	Observed identity / role	Key transit
R-BR	Branch router; neighbor 2.2.2.2 observed	10.10.252.1/30
R-ISP	Router-ID 2.2.2.2	10.10.252.2/30 and 10.10.253.2/30
R-HQ	HQ routed edge	10.10.253.1/30 and 10.10.254.1/30
SW-CORE	Router-ID 4.4.4.4	10.10.254.2/30 peer
R-INTERNET	Internet simulator	10.10.255.2/30

```
R-BR# show ip ospf neighbor
Neighbor ID State Address Interface
2.2.2.2 FULL/BDR 10.10.252.2 GigabitEthernet0/1
```

The route tables showed learned 10.10.x.x and 10.20.x.x networks. This matters because OSPF provides the internal reachability foundation that the later static default paths depend upon. OSPF is not the Internet default mechanism in this project; it is the dynamic interior protocol.

The troubleshooting record also showed why a routing protocol alone is not enough. Internal OSPF routes can be healthy while Internet-bound traffic still lacks a usable default route. The build therefore verifies both specific prefixes and the 0.0.0.0/0 path.

OSPF acceptance test

The minimum practical acceptance state is a FULL adjacency, valid learned routes in `show ip route`, and successful pings across the routed transit links.

OSPF verification

Check neighbor state first, then route installation. A FULL neighbor without the expected route indicates a network statement or topology issue; a missing neighbor indicates an adjacency problem.

Check	Expected evidence	If not correct
Configuration	Relevant command or endpoint screen matches this section	Return to the owning device/configuration
Behavior	Gateway, routing or service test passes	Trace the path one layer at a time
Record	Final state is saved and documented	Save the known-good state before further changes

Extended engineering note. The value of this section is not limited to the command itself. The project uses each feature as part of an observable service chain. A correct VLAN, for example, matters because it places the endpoint in the correct broadcast domain; a correct route matters because it moves the packet to the next control-plane boundary; a correct NAT rule matters because the return traffic must map back to the private source. The same evidence-first principle applies throughout the design.

For maintenance, compare the current state against the documented baseline before making changes. Capture the relevant verification output, make one controlled change, and repeat the test. This creates a simple audit trail and protects the last known-good state. In this lab, that discipline was especially useful for DHCP Snooping, AP uplink VLAN membership, default routes and NAT source coverage.

13 | Routing Verification

Connected, OSPF and default routes

Routing evidence

The final route model combines connected SVI/transit routes, OSPF-learned enterprise networks and explicit static defaults. This combination is what made both internal reachability and Internet testing predictable.

Device	Default route next hop	Purpose
SW-CORE	10.10.254.1	Send HQ Internet traffic to R-HQ
R-HQ	10.10.253.2	Send HQ edge traffic to R-ISP
R-ISP	10.10.255.2	Send Internet traffic to R-INTERNET
R-BR	10.10.252.2	Send branch Internet traffic to R-ISP

The final routing model is intentionally hybrid. Connected routes describe local VLANs and transit networks. OSPF describes remote enterprise destinations. Static defaults describe the northbound Internet exit at each boundary. This makes the routing table easier to reason about during troubleshooting.

During the build, R-BR already had a default route toward R-ISP. The Internet issue persisted because the edge path itself was incomplete. Once R-INTERNET was added and the R-ISP default was configured, branch Internet testing began to work. HQ still required its own default route chain through SW-CORE and R-HQ.

```
show ip route
show ip route 0.0.0.0
show ip ospf neighbor
ping 10.10.252.2
ping 10.10.253.2
ping 10.10.255.2
```

Default-path verification

Do not stop after proving internal reachability. The default route must exist at every Layer-3 boundary that is responsible for forwarding unknown destinations toward the Internet.

Check	Expected evidence	If not correct
Configuration	Relevant command or endpoint screen matches this section	Return to the owning device/configuration
Behavior	Gateway, routing or service test passes	Trace the path one layer at a time
Record	Final state is saved and documented	Save the known-good state before further changes

Extended engineering note. The value of this section is not limited to the command itself. The project uses each feature as part of an observable service chain. A correct VLAN, for example, matters because it places the endpoint in the correct broadcast domain; a correct route matters because it moves the packet to the next control-plane boundary; a correct NAT rule matters because the return traffic must map back to the private source. The same evidence-first principle applies throughout the design.

For maintenance, compare the current state against the documented baseline before making changes. Capture the relevant verification output, make one controlled change, and repeat the test. This creates a simple audit trail and protects the last known-good state. In this lab, that discipline was especially useful for DHCP Snooping, AP uplink VLAN membership, default routes and NAT source coverage.

14 | Branch DHCP Design

Scopes, exclusions and verified leases

R-BR is the authoritative DHCP server for the branch. Each pool matches a branch VLAN and supplies the appropriate .1 address as the default router. The configured DNS value for the lab is 8.8.8.8.

```
ip dhcp excluded-address 10.20.10.1 10.20.10.20
ip dhcp excluded-address 10.20.40.1 10.20.40.20
ip dhcp excluded-address 10.20.60.1 10.20.60.20
ip dhcp excluded-address 10.20.80.1 10.20.80.20
ip dhcp excluded-address 10.20.90.1 10.20.90.20
```

Pool	Network	Default gateway
BRANCH-USERS	10.20.10.0/24	10.20.10.1
BRANCH-IT	10.20.40.0/24	10.20.40.1
BRANCH-GUEST	10.20.60.0/24	10.20.60.1
BRANCH-IOT	10.20.80.0/24	10.20.80.1
BRANCH-WIFI	10.20.90.0/24	10.20.90.1

Verified leases during the final build included 10.20.10.21, 10.20.10.22, 10.20.10.24, 10.20.40.21, 10.20.60.22 and 10.20.90.22. Later, BR-WIFI-01 received 10.20.90.23, confirming that the Wi-Fi pool remained functional after the AP uplink was corrected.

DHCP troubleshooting rule

When a client has a 169.254.x.x APIPA address, do not immediately rebuild the DHCP pool. Check the client association, access VLAN, trunk path, router subinterface and DHCP service in that order.

DHCP verification

A valid pool is only one part of DHCP. Confirm the client VLAN, router subinterface, excluded addresses, lease table and client gateway together.

Check	Expected evidence	If not correct
Configuration	Relevant command or endpoint screen matches this section	Return to the owning device/configuration
Behavior	Gateway, routing or service test passes	Trace the path one layer at a time
Record	Final state is saved and documented	Save the known-good state before further changes

Extended engineering note. The value of this section is not limited to the command itself. The project uses each feature as part of an observable service chain. A correct VLAN, for example, matters because it places the endpoint in the correct broadcast domain; a correct route matters because it moves the packet to the next control-plane boundary; a correct NAT rule matters because the return traffic must map back to the private source. The same evidence-first principle applies throughout the design.

For maintenance, compare the current state against the documented baseline before making changes. Capture the relevant verification output, make one controlled change, and repeat the test. This creates a simple audit trail and protects the last known-good state. In this lab, that discipline was especially useful for DHCP Snooping, AP uplink VLAN membership, default routes and NAT source coverage.

15 | R-BR Configuration

Router-on-a-stick, branch gateways and default path

R-BR uses G0/0 as the physical trunk-facing interface and subinterfaces for each branch VLAN. The subinterfaces use 802.1Q encapsulation and provide the .1 gateway addresses from the branch addressing plan.

```
interface GigabitEthernet0/0.10
encapsulation dot1q 10
ip address 10.20.10.1 255.255.255.0
interface GigabitEthernet0/0.40
encapsulation dot1q 40
ip address 10.20.40.1 255.255.255.0
interface GigabitEthernet0/0.60
encapsulation dot1q 60
ip address 10.20.60.1 255.255.255.0
interface GigabitEthernet0/0.80
encapsulation dot1q 80
ip address 10.20.80.1 255.255.255.0
interface GigabitEthernet0/0.90
encapsulation dot1q 90
ip address 10.20.90.1 255.255.255.0
interface GigabitEthernet0/0.99
encapsulation dot1q 99
ip address 10.20.99.1 255.255.255.0
```

The WAN-facing G0/1 address is 10.10.252.1/30, with R-ISP at 10.10.252.2/30. The branch default route points to 10.10.252.2, while OSPF advertises the branch VLAN networks toward the rest of the enterprise.

```
ip route 0.0.0.0 0.0.0.0 10.10.252.2
router ospf 10
network 10.10.252.0 0.0.0.3 area 0
```

This arrangement centralizes branch gateway and DHCP functionality on one router while keeping the access switches focused on Layer-2 transport and security controls.

Branch gateway verification

Router-on-a-stick depends on the trunk carrying the VLAN ID that matches the subinterface encapsulation. A mismatch on either side breaks the client-to-gateway path.

Check	Expected evidence	If not correct
Configuration	Relevant command or endpoint screen matches this section	Return to the owning device/configuration
Behavior	Gateway, routing or service test passes	Trace the path one layer at a time
Record	Final state is saved and documented	Save the known-good state before further changes

Extended engineering note. The value of this section is not limited to the command itself. The project uses each feature as part of an observable service chain. A correct VLAN, for example, matters because it places the endpoint in the correct broadcast domain; a correct route matters because it moves the packet to the next control-plane boundary; a correct NAT rule matters because the return traffic must map back to the private source. The same evidence-first principle applies throughout the design.

For maintenance, compare the current state against the documented baseline before making changes. Capture the relevant verification output, make one controlled change, and repeat the test. This creates a simple audit trail and protects the last known-good state. In this lab, that discipline was especially useful for DHCP Snooping, AP uplink VLAN membership, default routes and NAT source coverage.

16 | R-HQ Configuration

HQ routed edge and explicit Internet default

R-HQ sits between SW-CORE and R-ISP. Its G0/0 interface uses 10.10.254.1/30 toward the core, while G0/1 uses 10.10.253.1/30 toward R-ISP. OSPF carries the internal routes in both directions.

```
interface GigabitEthernet0/0
ip address 10.10.254.1 255.255.255.252
interface GigabitEthernet0/1
ip address 10.10.253.1 255.255.255.252
ip route 0.0.0.0 0.0.0.0 10.10.253.2
```

The final HQ Internet path required a default route on both sides of the core boundary. SW-CORE sends unknown destinations toward 10.10.254.1, and R-HQ sends its own unknown destinations toward 10.10.253.2. Without the second route, R-HQ can know the internal networks perfectly well while still returning an unreachable result for Internet-bound traffic.

This illustrates a useful design principle: dynamic routing and Internet policy solve different problems. OSPF determines where known enterprise prefixes live. The static default determines where everything else should go.

Operational check

When HQ clients can ping 10.10.90.1 but receive destination-unreachable responses for 8.8.8.8, inspect `show ip route 0.0.0.0` on SW-CORE and R-HQ before changing the VLAN or DHCP design.

HQ edge verification

Treat R-HQ as a routing boundary. Verify its two transit links, OSPF-learned internal prefixes and explicit default before diagnosing NAT.

Check	Expected evidence	If not correct
Configuration	Relevant command or endpoint screen matches this section	Return to the owning device/configuration
Behavior	Gateway, routing or service test passes	Trace the path one layer at a time
Record	Final state is saved and documented	Save the known-good state before further changes

Extended engineering note. The value of this section is not limited to the command itself. The project uses each feature as part of an observable service chain. A correct VLAN, for example, matters because it places the endpoint in the correct broadcast domain; a correct route matters because it moves the packet to the next control-plane boundary; a correct NAT rule matters because the return traffic must map back to the private source. The same evidence-first principle applies throughout the design.

For maintenance, compare the current state against the documented baseline before making changes. Capture the relevant verification output, make one controlled change, and repeat the test. This creates a simple audit trail and protects the last known-good state. In this lab, that discipline was especially useful for DHCP Snooping, AP uplink VLAN membership, default routes and NAT source coverage.

17 | R-ISP Configuration

PAT edge, inside/outside roles and translation scope

R-ISP is the NAT/PAT boundary. G0/0 and G0/1 are inside interfaces facing R-HQ and R-BR. G0/2 is the outside interface toward R-INTERNET. The single outside IPv4 address is shared by many private source sessions.

```
interface GigabitEthernet0/0
ip address 10.10.253.2 255.255.255.252
ip nat inside
interface GigabitEthernet0/1
ip address 10.10.252.2 255.255.255.252
ip nat inside
interface GigabitEthernet0/2
ip address 10.10.255.1 255.255.255.252
ip nat outside
```

The translation policy uses ACL 1. The final ACL permits both private enterprise supernets so that branch and HQ traffic are eligible for PAT.

```
access-list 1 permit 10.20.0.0 0.0.255.255
access-list 1 permit 10.10.0.0 0.0.255.255
ip nat inside source list 1 interface GigabitEthernet0/2 overload
ip route 0.0.0.0 0.0.0.0 10.10.255.2
```

The word `overload` is the key to PAT behavior: multiple inside-local addresses and ports can share the one inside-global address on G0/2. The build demonstrated active translations only after client traffic was generated, which is why an empty table at idle time was not treated as a failure.

NAT verification

Inside/outside roles, source ACL coverage and the outside next hop must all agree. Generate traffic before inspecting translations so the table contains a real session.

Check	Expected evidence	If not correct
Configuration	Relevant command or endpoint screen matches this section	Return to the owning device/configuration
Behavior	Gateway, routing or service test passes	Trace the path one layer at a time
Record	Final state is saved and documented	Save the known-good state before further changes

Extended engineering note. The value of this section is not limited to the command itself. The project uses each feature as part of an observable service chain. A correct VLAN, for example, matters because it places the endpoint in the correct broadcast domain; a correct route matters because it moves the packet to the next control-plane boundary; a correct NAT rule matters because the return traffic must map back to the private source. The same evidence-first principle applies throughout the design.

For maintenance, compare the current state against the documented baseline before making changes. Capture the relevant verification output, make one controlled change, and repeat the test. This creates a simple audit trail and protects the last known-good state. In this lab, that discipline was especially useful for DHCP Snooping, AP uplink VLAN membership, default routes and NAT source coverage.

18 | R-INTERNET Configuration

Self-contained Internet simulation and test endpoint

R-INTERNET exists specifically to keep the Packet Tracer project self-contained. It is not a realistic commercial ISP router; it is a minimal next-hop device with an IP interface toward R-ISP and a loopback address that behaves like a stable public test destination.

```
interface GigabitEthernet0/0
ip address 10.10.255.2 255.255.255.252
no shutdown
interface Loopback0
ip address 8.8.8.8 255.255.255.255
```

The R-ISP side uses 10.10.255.1/30 as the NAT outside address and forwards unknown destinations to 10.10.255.2. The loopback remains independent of physical interface state, so it provides a reliable test target even though the project has no actual upstream Internet.

Validation demonstrated that R-ISP could ping 10.10.255.2 and 8.8.8.8. After PAT and the relevant default routes were corrected, client-originated traffic also reached the same endpoint through the full enterprise path.

Test	Expected observation
R-ISP -> 10.10.255.2	100% ICMP reachability
R-ISP -> 8.8.8.8	100% ICMP reachability
Client -> 8.8.8.8	Reply after routing + NAT are complete
R-ISP -> show ip nat translations	Dynamic entries appear after client traffic

Scope limitation

The 8.8.8.8 endpoint validates routing, PAT and return traffic. It does not create a real public DNS service or real Internet connectivity.

Internet simulator verification

Use 8.8.8.8 as the acceptance endpoint. A successful ICMP result demonstrates the simulated path; it does not imply public DNS or real Internet service.

Check	Expected evidence	If not correct
Configuration	Relevant command or endpoint screen matches this section	Return to the owning device/configuration
Behavior	Gateway, routing or service test passes	Trace the path one layer at a time
Record	Final state is saved and documented	Save the known-good state before further changes

Extended engineering note. The value of this section is not limited to the command itself. The project uses each feature as part of an observable service chain. A correct VLAN, for example, matters because it places the endpoint in the correct broadcast domain; a correct route matters because it moves the packet to the next control-plane boundary; a correct NAT rule matters because the return traffic must map back to the private source. The same evidence-first principle applies throughout the design.

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19 | Trunk Configuration Standard

Allowed VLANs, native VLAN and troubleshooting value

The branch trunk design explicitly allows the production VLANs 10, 40, 60, 80, 90 and 99. The native VLAN remains VLAN 1. This is a deliberate standard: the trunk carries only the VLANs required by the project and does not rely on an unrestricted allowed list.

```
interface Port-channel4
switchport trunk allowed vlan 10,40,60,80,90,99
switchport mode trunk
interface GigabitEthernet0/1
switchport trunk allowed vlan 10,40,60,80,90,99
switchport mode trunk
```

`show interfaces trunk` was repeatedly useful because it simultaneously exposes the trunking state, native VLAN, allowed VLANs and VLANs that are active and forwarding. This is more informative during DHCP troubleshooting than looking only at an endpoint IP address.

Troubleshooting symptom	First trunk question
Client has APIPA	Is the client VLAN permitted on every required trunk?
AP has link but no DHCP	Does the AP access VLAN match the intended service VLAN?
Remote VLAN unreachable	Is the VLAN active and forwarding across the port-channel?
Inter-switch instability	Is the EtherChannel formed rather than two independent trunks?

The final build benefited from this exact sequence: identify the endpoint VLAN, verify the local access port, verify the trunk path, and only then inspect the router subinterface or DHCP service.

Trunk troubleshooting method

When a VLAN service fails, compare the local access VLAN, the trunk allowed list and the router subinterface VLAN ID. This ordered comparison isolates most Layer-2/Layer-3 handoff errors.

Check	Expected evidence	If not correct
Configuration	Relevant command or endpoint screen matches this section	Return to the owning device/configuration
Behavior	Gateway, routing or service test passes	Trace the path one layer at a time
Record	Final state is saved and documented	Save the known-good state before further changes

Extended engineering note. The value of this section is not limited to the command itself. The project uses each feature as part of an observable service chain. A correct VLAN, for example, matters because it places the endpoint in the correct broadcast domain; a correct route matters because it moves the packet to the next control-plane boundary; a correct NAT rule matters because the return traffic must map back to the private source. The same evidence-first principle applies throughout the design.

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20 | Guest Network Isolation

ACL-GUEST-IN and least-privilege policy

Guest traffic is sourced from the HQ GUEST VLAN 60, 10.10.60.0/24. The policy is applied inbound on the VLAN 60 SVI at SW-CORE. Specific deny statements block direct access to the enterprise 10.10.x.x subnets, followed by a general permit for other destinations.

```
ip access-list extended ACL-GUEST-IN
deny ip 10.10.60.0 0.0.0.255 10.10.10.0 0.0.0.255
deny ip 10.10.60.0 0.0.0.255 10.10.20.0 0.0.0.255
deny ip 10.10.60.0 0.0.0.255 10.10.30.0 0.0.0.255
deny ip 10.10.60.0 0.0.0.255 10.10.40.0 0.0.0.255
deny ip 10.10.60.0 0.0.0.255 10.10.50.0 0.0.0.255
deny ip 10.10.60.0 0.0.0.255 10.10.70.0 0.0.0.255
deny ip 10.10.60.0 0.0.0.255 10.10.80.0 0.0.0.255
deny ip 10.10.60.0 0.0.0.255 10.10.90.0 0.0.0.255
deny ip 10.10.60.0 0.0.0.255 10.10.99.0 0.0.0.255
permit ip 10.10.60.0 0.0.0.255 any
```

The validation tests intentionally targeted internal gateway addresses such as 10.10.10.1, 10.10.20.1 and 10.10.70.1. Those pings failed from the Guest segment as expected, demonstrating that the access boundary was functioning.

Security interpretation

This is a least-privilege boundary: Guest users are not automatically trusted simply because routing exists. Traffic is permitted only after explicitly excluded enterprise destinations have been denied.

Guest policy verification

Test from the Guest source network toward each protected enterprise subnet. Policy validation should include both a blocked internal destination and an allowed external destination.

Check	Expected evidence	If not correct
Configuration	Relevant command or endpoint screen matches this section	Return to the owning device/configuration
Behavior	Gateway, routing or service test passes	Trace the path one layer at a time
Record	Final state is saved and documented	Save the known-good state before further changes

Extended engineering note. The value of this section is not limited to the command itself. The project uses each feature as part of an observable service chain. A correct VLAN, for example, matters because it places the endpoint in the correct broadcast domain; a correct route matters because it moves the packet to the next control-plane boundary; a correct NAT rule matters because the return traffic must map back to the private source. The same evidence-first principle applies throughout the design.

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21 | SSH Management Security

SSH version 2, local authentication and management access

The management plane uses SSH version 2 rather than Telnet. Devices use a local administrative account and VTY lines configured for local login with SSH as the allowed transport. The project also uses the domain name corp.local and a login banner.

```
ip ssh version 2
ip domain-name corp.local
username admin privilege 15 secret &lt;lab-secret>
line vty 0 4
login local
transport input ssh
line vty 5 15
login local
transport input ssh
banner motd ^CAUTHORIZED ACCESS ONLY^C
```

This configuration provides confidentiality for remote management sessions and removes Telnet from the normal remote-administration path. The domain name also supports SSH key generation in the Cisco IOS workflow.

Verification used `show ip ssh`, which reported SSH Enabled - version 2.0. The project documentation intentionally does not reproduce actual password values; it records the authentication mechanism and account role instead.

Management control	Final implementation
Protocol	SSH version 2
Authentication	Local user database
Privilege	Admin account at privilege 15
Remote transport	SSH only on configured VTY lines
Warning	AUTHORIZED ACCESS ONLY banner

SSH verification

Confirm SSH version 2, local authentication and SSH-only VTY transport. Management controls should be tested from a trusted management source rather than from an ordinary user endpoint.

Check	Expected evidence	If not correct
Configuration	Relevant command or endpoint screen matches this section	Return to the owning device/configuration
Behavior	Gateway, routing or service test passes	Trace the path one layer at a time
Record	Final state is saved and documented	Save the known-good state before further changes

Extended engineering note. The value of this section is not limited to the command itself. The project uses each feature as part of an observable service chain. A correct VLAN, for example, matters because it places the endpoint in the correct broadcast domain; a correct route matters because it moves the packet to the next control-plane boundary; a correct NAT rule matters because the return traffic must map back to the private source. The same evidence-first principle applies throughout the design.

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22 | Port Security

Sticky MAC learning, Restrict mode and recovery lesson

Host-facing branch access ports use Port Security with sticky MAC learning. Restrict mode is used for violations, meaning unauthorized frames are blocked and the violation counter increases without immediately taking the interface down.

```
interface FastEthernet0/1
switchport mode access
switchport port-security
switchport port-security mac-address sticky
switchport port-security violation restrict
spanning-tree portfast
spanning-tree bpduguard enable
```

Switch	Port	VLAN	Security state
SW-BR1	Fa0/1	10	Sticky, restrict
SW-BR1	Fa0/2	10	Sticky, restrict
SW-BR1	Fa0/3	40	Sticky, restrict
SW-BR2	Fa0/1	10	Sticky, restrict
SW-BR2	Fa0/2	10	Sticky, restrict
SW-BR2	Fa0/3	60	Sticky, restrict
SW-BR2	Fa0/4	90	Sticky, restrict

One of the troubleshooting events involved an incorrect sticky MAC association on SW-BR2 Fa0/1. The stale manual sticky entry was removed so the expected endpoint could use the correct access-port association. The lesson is operational: sticky MAC state is useful, but stale learned state must be managed during topology changes.

Verification

Use `show port-security` to confirm the maximum secure addresses, current secure addresses, violation count and violation action for the relevant access ports.

Port-security verification

Record the expected secure MAC count and violation action before moving a device. If hardware or topology changes, clear or correct stale sticky state deliberately.

Check	Expected evidence	If not correct
Configuration	Relevant command or endpoint screen matches this section	Return to the owning device/configuration
Behavior	Gateway, routing or service test passes	Trace the path one layer at a time
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Extended engineering note. The value of this section is not limited to the command itself. The project uses each feature as part of an observable service chain. A correct VLAN, for example, matters because it places the endpoint in the correct broadcast domain; a correct route matters because it moves the packet to the next control-plane boundary; a correct NAT rule matters because the return traffic must map back to the private source. The same evidence-first principle applies throughout the design.

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23 | DHCP Snooping

Implementation, rate limiting and Packet Tracer exception

DHCP Snooping was enabled on branch switches for VLANs 10, 40, 60, 80 and 99. Selected access ports used a 10 packets-per-second limit, while trusted uplinks were configured where supported.

During branch Wi-Fi testing, VLAN 90 on SW-BR2 produced repeated DHCP-rate-limit err-disable messages on Fa0/4. The observed log sequence was `DHCP\_SNOOPING\_ERRDISABLE\_WARNING` followed by `ERR\_DISABLE: dhcp-rate-limit error detected on Fa0/4`.

```
show ip dhcp snooping
Switch DHCP snooping is disabled
DHCP snooping is configured on following VLANs:
10,40,60,80,99
```

The final stable state therefore disabled DHCP Snooping globally on SW-BR2 while retaining configuration lines for the other VLANs. SW-BR1 retained snooping for VLANs 10, 40, 60, 80 and 99. This was a Packet Tracer compatibility workaround rather than the preferred production pattern.

Preferred enterprise behavior	Final lab workaround
Trust only legitimate DHCP-server/uplink paths	Disable unstable snooping on SW-BR2
Rate-limit hostile DHCP behavior	Do not re-enable VLAN 90 until simulator behavior is re-tested
Build bindings for DAI	Keep working services stable for the lab acceptance test

Important caveat

The project must not be described as using production-grade DHCP Snooping uniformly across all branch VLANs. The exact compatibility exception above is part of the final record.

DHCP Snooping exception

The final SW-BR2 state is a simulator workaround. Do not present it as a recommended production configuration without first reproducing the behavior on the target platform.

Check	Expected evidence	If not correct
Configuration	Relevant command or endpoint screen matches this section	Return to the owning device/configuration
Behavior	Gateway, routing or service test passes	Trace the path one layer at a time
Record	Final state is saved and documented	Save the known-good state before further changes

Extended engineering note. The value of this section is not limited to the command itself. The project uses each feature as part of an observable service chain. A correct VLAN, for example, matters because it places the endpoint in the correct broadcast domain; a correct route matters because it moves the packet to the next control-plane boundary; a correct NAT rule matters because the return traffic must map back to the private source. The same evidence-first principle applies throughout the design.

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24 | Dynamic ARP Inspection

ARP-spoofing protection and trust model

Dynamic ARP Inspection is enabled for VLANs 10, 40, 60, 80, 90 and 99 on the access switches. Uplinks and trunk members were made trusted where the Packet Tracer platform accepted the configuration, while ordinary access ports remain untrusted.

```
ip arp inspection vlan 10,40,60,80,90,99
show ip arp inspection
Vlan 10 Enabled Active
Vlan 40 Enabled Active
Vlan 60 Enabled Active
Vlan 80 Enabled Active
Vlan 90 Enabled Active
Vlan 99 Enabled Active
```

The observed validation counters remained at zero drops and zero validation failures while legitimate endpoints were tested. That is the desired result: the control should remain active without disrupting valid addressing.

DAI complements DHCP Snooping rather than replacing it. In a production design, the switch uses trusted DHCP bindings as a source of truth for ARP validation. The Packet Tracer exception on SW-BR2 is therefore relevant: the lab kept DAI enabled but adjusted DHCP Snooping behavior where the simulator was unstable.

Security takeaway

DAI is most effective when trust is carefully scoped. The final topology distinguishes trusted trunks/uplinks from untrusted client-facing ports, which is the correct conceptual model for an enterprise access layer.

DAI verification

Confirm the VLAN is Active under DAI and inspect trusted uplinks versus untrusted access ports. Keep DAI policy aligned with the binding source available on the platform.

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Configuration	Relevant command or endpoint screen matches this section	Return to the owning device/configuration
Behavior	Gateway, routing or service test passes	Trace the path one layer at a time
Record	Final state is saved and documented	Save the known-good state before further changes

Extended engineering note. The value of this section is not limited to the command itself. The project uses each feature as part of an observable service chain. A correct VLAN, for example, matters because it places the endpoint in the correct broadcast domain; a correct route matters because it moves the packet to the next control-plane boundary; a correct NAT rule matters because the return traffic must map back to the private source. The same evidence-first principle applies throughout the design.

For maintenance, compare the current state against the documented baseline before making changes. Capture the relevant verification output, make one controlled change, and repeat the test. This creates a simple audit trail and protects the last known-good state. In this lab, that discipline was especially useful for DHCP Snooping, AP uplink VLAN membership, default routes and NAT source coverage.

25 | HQ Wi-Fi Deployment

AP-HQ, WPC300N clients and VLAN 90

HQ Wi-Fi evidence

AP-HQ was configured with HQ-WIFI, WPA2-PSK, AES and channel 6. Both HQ wireless clients associated successfully and obtained the documented 10.10.90.x addresses.

AP-HQ uses SSID `HQ-WIFI`, WPA2-PSK, AES encryption and channel 6. HQ-WIFI-LAP01 and HQ-WIFI-LAP02 use WPC300N wireless adapters. Both clients successfully associated and obtained addresses from the HQ Wi-Fi network.

Client	IPv4	Mask	Gateway	Validation
HQ-WIFI-LAP01	10.10.90.21	255.255.255.0	10.10.90.1	Gateway + Internet OK
HQ-WIFI-LAP02	10.10.90.22	255.255.255.0	10.10.90.1	Gateway + Internet OK after route/NAT fix

The wireless clients are deliberately placed in VLAN 90, not VLAN 80. This keeps user Wi-Fi separate from the IoT service. The AP profile and switch access VLAN must agree with the addressing plan; once those dependencies were correct, DHCP and Internet validation completed normally.

Client workflow

Install the WPC300N adapter if the endpoint is a desktop-style Packet Tracer device, associate to the SSID using WPA2-Personal and the PSK, then set IPv4 to DHCP. Verify both the gateway and the Internet endpoint.

HQ Wi-Fi onboarding

For each client, verify wireless association first, then DHCP, then the gateway, then Internet reachability. The WPC300N adapter is required by the Packet Tracer endpoint used here.

Check	Expected evidence	If not correct
Configuration	Relevant command or endpoint screen matches this section	Return to the owning device/configuration
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26 | Branch Wi-Fi Deployment

AP-BR, VLAN 90 and the successful client leases

Branch Wi-Fi evidence

AP-BR was configured with BR-WIFI, WPA2-PSK, AES and channel 11. Its wired uplink was placed in VLAN 90 on SW-BR2 Fa0/10 before the two branch wireless clients could obtain their final leases.

AP-BR uses SSID `BR-WIFI`, WPA2-PSK, AES encryption and channel 11. Its final wired uplink is SW-BR2 Fa0/10, configured as an access port in VLAN 90. That switchport assignment was the key correction that restored the service.

Client	IPv4	Gateway	Internet
BR-WIFI-01	10.20.90.23	10.20.90.1	8.8.8.8 successful
BR-WIFI-02	10.20.90.22	10.20.90.1	8.8.8.8 successful

Before the fix, both BR Wi-Fi endpoints received 169.254.x.x APIPA addresses. The DHCP service on R-BR itself was healthy. The actual issue was the AP uplink entering VLAN 1 rather than VLAN 90. Once Fa0/10 was corrected to VLAN 90, the DHCP Discover traffic reached the G0/0.90 router subinterface and valid addresses were assigned.

```
interface FastEthernet0/10
switchport mode access
switchport access vlan 90
no shutdown
```

Branch Wi-Fi onboarding

The wireless profile and AP switchport form one service. The AP must be in VLAN 90 and the branch client must receive a 10.20.90.x address.

Check	Expected evidence	If not correct
Configuration	Relevant command or endpoint screen matches this section	Return to the owning device/configuration
Behavior	Gateway, routing or service test passes	Trace the path one layer at a time
Record	Final state is saved and documented	Save the known-good state before further changes

Extended engineering note. The value of this section is not limited to the command itself. The project uses each feature as part of an observable service chain. A correct VLAN, for example, matters because it places the endpoint in the correct broadcast domain; a correct route matters because it moves the packet to the next control-plane boundary; a correct NAT rule matters because the return traffic must map back to the private source. The same evidence-first principle applies throughout the design.

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27 | IoT Wireless Deployment

AP-IOT, VLAN 80 and three IoT endpoints

IoT evidence

AP-IOT uses HQ-IOT with WPA2-PSK/AES on channel 6 and is connected to SW-CORE Fa0/9 in VLAN 80. All three IoT endpoints received 10.10.80.x addresses.

AP-IOT is connected to SW-CORE Fa0/9, and Fa0/9 is an access port in VLAN 80. The wireless profile uses SSID `HQ-IOT`, WPA2-PSK, AES encryption and channel 6. The service is intentionally separate from HQ user Wi-Fi.

IoT endpoint	IPv4	Mask	Gateway
IOT-TEMP01	10.10.80.22	255.255.255.0	10.10.80.1
IOT-MOTION01	10.10.80.23	255.255.255.0	10.10.80.1
IOT-DOOR01	10.10.80.24	255.255.255.0	10.10.80.1

Unlike PCs and laptops, the Packet Tracer IoT device types used in this project expose their IPv4 state directly through the Wireless0 configuration page. Expecting a Desktop tab on those devices created unnecessary confusion during the build. The final record explicitly documents this user-interface distinction so the implementation can be reproduced.

The separate IoT VLAN also provides a stronger policy boundary. If an ACL later needs to restrict sensor access to one server, the source network is already isolated as 10.10.80.0/24.

IoT onboarding

Keep IoT isolated in VLAN 80. For Packet Tracer IoT devices, inspect Wireless0 directly because the interface differs from a PC or laptop Desktop/IP Configuration workflow.

Check	Expected evidence	If not correct
Configuration	Relevant command or endpoint screen matches this section	Return to the owning device/configuration
Behavior	Gateway, routing or service test passes	Trace the path one layer at a time
Record	Final state is saved and documented	Save the known-good state before further changes

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28 | Wireless Security Standards

SSID, encryption, channel and VLAN policy

AP	SSID	Purpose	Security	Channel	VLAN
AP-HQ	HQ-WIFI	HQ user Wi-Fi	WPA2-PSK / AES	6	90
AP-BR	BR-WIFI	Branch user Wi-Fi	WPA2-PSK / AES	11	90
AP-IOT	HQ-IOT	HQ IoT devices	WPA2-PSK / AES	6	80

The security pattern is intentionally consistent: WPA2-PSK and AES are used for all three APs, while the SSID and VLAN identify the service class. User Wi-Fi and IoT do not share a broadcast domain, even though they use similar security settings.

Channel selection in this Packet Tracer lab is primarily an organizational configuration rather than an RF optimization study. AP-HQ and AP-IOT both use channel 6 because they are separate service networks in the simulation; AP-BR uses channel 11. In a physical deployment, channel planning would additionally consider neighboring access points, interference, power and client density.

The final wireless acceptance criteria are therefore simple and observable: the client sees the intended SSID, completes WPA2 authentication, receives the expected DHCP subnet, reaches its gateway, and reaches the simulated Internet when the path is permitted.

Naming convention

`HQ-WIFI` identifies user wireless at headquarters, `BR-WIFI` identifies user wireless at the branch, and `HQ-IOT` identifies the dedicated IoT service. Consistent naming reduces operator ambiguity during endpoint onboarding.

Wireless security interpretation

User Wi-Fi and IoT can share the same security technology while remaining logically isolated by SSID and VLAN. In physical deployments, RF planning would add another design layer.

Check	Expected evidence	If not correct
Configuration	Relevant command or endpoint screen matches this section	Return to the owning device/configuration
Behavior	Gateway, routing or service test passes	Trace the path one layer at a time
Record	Final state is saved and documented	Save the known-good state before further changes

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29 | Troubleshooting Record: Branch DHCP Failure

APIPA symptoms, err-disable evidence and resolution path

The first major branch incident was a DHCP failure on BR-PC04. The client initially showed 169.254.3.202 and repeated DHCP request failures. The correct troubleshooting approach was to walk the packet path from the endpoint outward rather than repeatedly changing the DHCP pool.

```
DHCP_SNOOPING_ERRDISABLE_WARNING: DHCP Snooping received 1 DHCP packets on interface Fa0/4
ERR_DISABLE: dhcp-rate-limit error detected on Fa0/4
Interface FastEthernet0/4, changed state to down
```

This log sequence proved that DHCP Discover traffic was reaching the switch, but the switch security feature was reacting to it in a way that repeatedly put the interface into err-disabled state. The rate limit was removed during recovery, the interface was bounced, and the final stable state disabled DHCP Snooping globally on SW-BR2.

Once Fa0/4 remained available, BR-PC04 obtained 10.20.90.41. The branch Wi-Fi service later became stable as well. The troubleshooting lesson is important: when an endpoint is receiving APIPA, distinguish between DHCP-server failure and a switch security feature that is preventing the request from being forwarded.

Evidence	Interpretation
169.254.x.x	No usable DHCP lease
DHCP Snooping warning	Switch sees DHCP on the interface
ERR_DISABLE message	Interface is being shut due to the rate-limit behavior
Valid 10.20.90.x lease after recovery	DHCP service and router subinterface are healthy

Incident timeline

The branch DHCP incident was solved by using evidence from the switch logs instead of repeatedly changing the router DHCP pool. This is a model troubleshooting pattern for enterprise operations.

Check	Expected evidence	If not correct
Configuration	Relevant command or endpoint screen matches this section	Return to the owning device/configuration
Behavior	Gateway, routing or service test passes	Trace the path one layer at a time
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30 | Troubleshooting Record: AP-BR VLAN Error

Finding VLAN 1 instead of VLAN 90

After the DHCP Snooping issue was stabilized, the branch Wi-Fi laptops still received APIPA addresses. At that point the correct question was no longer “is the DHCP pool right?” but “does the AP uplink actually belong to the wireless VLAN?”

The switch VLAN table showed that Fa0/10, the physical AP-BR uplink, was still a member of VLAN 1. VLAN 90 was active on the switch, but the AP traffic was arriving in the wrong broadcast domain.

State	Fa0/10 membership
Before fix	VLAN 1
Required state	VLAN 90
Final validated state	Connected, VLAN 90

```
configure terminal
interface fastethernet0/10
switchport mode access
switchport access vlan 90
no shutdown
end
write memory
```

After the correction, BR-WIFI-01 obtained 10.20.90.23 and BR-WIFI-02 obtained 10.20.90.22. Both reached 10.20.90.1 and 8.8.8.8 successfully. This incident reinforces an important enterprise principle: the AP profile and the wired switchport configuration form one service. One without the other is not a functioning wireless deployment.

AP VLAN incident timeline

The AP-BR problem demonstrates that wireless configuration cannot be validated in isolation. The SSID, client authentication, AP uplink VLAN and router subinterface must all agree.

Check	Expected evidence	If not correct
Configuration	Relevant command or endpoint screen matches this section	Return to the owning device/configuration
Behavior	Gateway, routing or service test passes	Trace the path one layer at a time
Record	Final state is saved and documented	Save the known-good state before further changes

Extended engineering note. The value of this section is not limited to the command itself. The project uses each feature as part of an observable service chain. A correct VLAN, for example, matters because it places the endpoint in the correct broadcast domain; a correct route matters because it moves the packet to the next control-plane boundary; a correct NAT rule matters because the return traffic must map back to the private source. The same evidence-first principle applies throughout the design.

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31 | Troubleshooting Record: HQ Internet Reachability

Default route chain and NAT source coverage

HQ Wi-Fi initially reached 10.10.90.1 successfully but received destination-unreachable results when testing 8.8.8.8. Because branch Internet traffic already worked, the investigation compared the forwarding chains instead of rebuilding common services.

Hop	Required result
HQ client	10.10.90.x with gateway 10.10.90.1
SW-CORE	Default route to 10.10.254.1
R-HQ	Default route to 10.10.253.2
R-ISP	PAT eligibility for 10.10.0.0/16 plus 10.20.0.0/16
R-INTERNET	10.10.255.2 reachable and 8.8.8.8 loopback active

```
SW-CORE(config)# ip route 0.0.0.0 0.0.0.0 10.10.254.1
R-HQ(config)# ip route 0.0.0.0 0.0.0.0 10.10.253.2
R-ISP(config)# access-list 1 permit 10.10.0.0 0.0.255.255
```

The NAT ACL expansion was the final missing detail for HQ Wi-Fi sources. Branch traffic already matched 10.20.0.0/16, but HQ clients used 10.10.90.x and therefore were not translated until the 10.10.0.0/16 entry was added.

Root-cause sequence

The incident was not one failure. It was a dependency chain: missing core default -> missing R-HQ default -> missing HQ NAT source coverage. Each correction was validated before moving to the next layer.

HQ Internet incident timeline

The issue was a chain of missing dependencies, not a single command. Route outward from the client through each Layer-3 hop and verify the next hop before moving farther.

Check	Expected evidence	If not correct
Configuration	Relevant command or endpoint screen matches this section	Return to the owning device/configuration
Behavior	Gateway, routing or service test passes	Trace the path one layer at a time
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32 | NAT/PAT Validation

Dynamic translations prove the Internet path

NAT evidence

The active translation table contained private inside-local sources mapped to 10.10.255.1 with 8.8.8.8 as the outside destination. This demonstrates that PAT was active during real traffic.

The R-ISP NAT table showed inside-local addresses from the enterprise networks being translated to 10.10.255.1 as the inside-global address. The outside destination was 8.8.8.8, the simulated Internet endpoint.

Command	Purpose
show ip nat translations	Displays active local/global address and port mappings
show ip nat statistics	Shows hit/miss counters and inside/outside interfaces
ping 8.8.8.8 from a client	Creates real traffic that should generate a translation

A useful operational point from the build is that the NAT table may be empty between traffic events. The correct acceptance method is therefore to generate client traffic and immediately inspect the table. During the final test, translations appeared for the validated branch and HQ Wi-Fi sessions.

The PAT design is simple enough to explain in an interview: inside enterprise subnets are matched by ACL 1, all matching sessions use the single public-facing address on G0/2, and the router tracks source port information so several clients can share the same outside address.

PAT evidence

An active translation is stronger evidence than a configured NAT command. Create traffic from more than one source when possible to demonstrate that PAT can multiplex sessions.

Check	Expected evidence	If not correct
Configuration	Relevant command or endpoint screen matches this section	Return to the owning device/configuration
Behavior	Gateway, routing or service test passes	Trace the path one layer at a time
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33 | Server and DNS Considerations

VLAN 70, SRV-DNS2 and simulator limits

VLAN 70 is reserved for SERVER-DMZ at HQ. The topology includes a server labeled SRV-DNS2. The project uses 8.8.8.8 as the DNS server value in DHCP configuration, but the R-INTERNET loopback at 8.8.8.8 is fundamentally an ICMP test endpoint in this lab.

This distinction matters. A successful `ping 8.8.8.8` proves IP reachability and return traffic. It does not guarantee that `google.com` will resolve. During the build, a name-resolution test could fail even while the Internet-path test was healthy because the simulated endpoint is not a complete public DNS implementation.

Test	What it proves
ping 10.10.70.x	Reachability to a service in the Server-DMZ segment if permitted
ping 8.8.8.8	Routing + NAT + return path to the simulated Internet endpoint
ping google.com	Requires an actual DNS resolver with that record; not guaranteed by this lab

A production-like extension would place an internal DNS service in VLAN 70 or a dedicated services segment, update DHCP pools to reference that internal resolver, and protect access to DNS with explicit ACL rules.

Documentation boundary

Do not present the simulated 8.8.8.8 address as a real public resolver. It is the acceptance endpoint for the Packet Tracer Internet simulation.

DNS boundary

Treat the configured DNS address as a lab value and document the simulator limitation. A real production implementation should separate recursive DNS service from the ICMP-only test endpoint.

Check	Expected evidence	If not correct
Configuration	Relevant command or endpoint screen matches this section	Return to the owning device/configuration
Behavior	Gateway, routing or service test passes	Trace the path one layer at a time
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34 | Management Plane Design

VLAN 99, administrative separation and future AAA

Management VLAN 99 is a dedicated administrative control-plane segment. HQ uses 10.10.99.0/24 and the branch uses 10.20.99.0/24. Management addressing is kept separate from normal user and wireless traffic so that future access-class, ACL or AAA controls can be added without redesigning the user subnets.

Control	Implemented state
SSH	Version 2 only on configured VTY lines
Authentication	Local admin account, privilege 15
Domain	corp.local
Banner	AUTHORIZED ACCESS ONLY
Telnet	Not permitted on configured VTY lines
Unused ports	Administratively shut only where disconnected status was known

The project does not include a full TACACS+ or RADIUS deployment. That is a scope boundary rather than a design flaw: the VLAN and SSH structure is already prepared for centralized AAA to be added later.

From an operational perspective, the management plane should be the first area to protect when extending the lab. Management access should be limited to trusted administrative endpoints, logging should be centralized, and credentials should be externalized to AAA when the environment grows beyond a training lab.

Interview point

A strong enterprise explanation is: “User traffic is segmented, management traffic has its own subnet, remote administration uses SSHv2, and the design leaves room for AAA without changing the IP plan.”

Management extension path

The current design can be extended with centralized AAA, management ACLs, NTP, Syslog and SNMP without changing the basic VLAN plan. Build those controls from the dedicated management segment.

Check	Expected evidence	If not correct
Configuration	Relevant command or endpoint screen matches this section	Return to the owning device/configuration
Behavior	Gateway, routing or service test passes	Trace the path one layer at a time
Record	Final state is saved and documented	Save the known-good state before further changes

Extended engineering note. The value of this section is not limited to the command itself. The project uses each feature as part of an observable service chain. A correct VLAN, for example, matters because it places the endpoint in the correct broadcast domain; a correct route matters because it moves the packet to the next control-plane boundary; a correct NAT rule matters because the return traffic must map back to the private source. The same evidence-first principle applies throughout the design.

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35 | Endpoint Test Results

Representative final client outcomes

Endpoint	Address	Gateway test	Internet test
HQ-WIFI-LAP01	10.10.90.21	PASS	PASS
HQ-WIFI-LAP02	10.10.90.22	PASS	PASS after route/NAT fix
BR-WIFI-01	10.20.90.23	PASS	PASS
BR-WIFI-02	10.20.90.22	PASS	PASS
IOT-TEMP01	10.10.80.22	10.10.80.1 expected	Addressing validated
IOT-MOTION01	10.10.80.23	10.10.80.1 expected	Addressing validated
IOT-DOOR01	10.10.80.24	10.10.80.1 expected	Addressing validated

The testing model uses two levels. A gateway test isolates Layer-2, access-port, VLAN and local gateway problems. The 8.8.8.8 test then checks the longer path: routing, PAT and return reachability. This method proved faster and more precise than relying on one end-to-end ping as the only test.

The final verified wireless addresses are especially useful because they map cleanly to the design: 10.10.90.x means HQ Wi-Fi, 10.20.90.x means branch Wi-Fi, and 10.10.80.x means HQ IoT.

Validation evidence

Final endpoint and gateway tests agreed with the addressing plan. The acceptance matrix below is the authoritative summary of those checks.

Validation discipline

Keep a copy of the final endpoint table. Repeating the same tests after any change makes regression visible and prevents a “works now but breaks another service” outcome.

Check	Expected evidence	If not correct
Configuration	Relevant command or endpoint screen matches this section	Return to the owning device/configuration
Behavior	Gateway, routing or service test passes	Trace the path one layer at a time
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36 | Verification Matrix

What was checked before declaring the project complete

Area	Verification command / method	Expected result	Status
VLAN	show vlan brief	Correct access membership	PASS
Trunk	show interfaces trunk	Required VLANs allowed/forwarding	PASS
LACP	show etherchannel summary	Po4 with LACP members	PASS
OSPF	show ip ospf neighbor	FULL adjacency	PASS
DHCP	show ip dhcp binding	Client leases present	PASS
Guest ACL	Ping internal HQ from Guest	Blocked	PASS
HQ Wi-Fi	Client addressing	10.10.90.x	PASS
Branch Wi-Fi	Client addressing	10.20.90.x	PASS
IoT	Device addressing	10.10.80.x	PASS
NAT	show ip nat translations	Dynamic mappings after traffic	PASS
Internet	Ping 8.8.8.8	Replies	PASS

The only documented caveat is the SW-BR2 DHCP Snooping compatibility workaround. It was retained because the final objective was a stable and repeatable Packet Tracer implementation and the branch Wi-Fi service depended on the workaround.

Acceptance statement

All major planned services were observed in operation at the end of the build. The project is therefore suitable as a complete enterprise networking lab, portfolio project and interview demonstration.

For formal handover, the verification matrix should be kept with the final Packet Tracer `.pkt` file so that a reviewer can repeat the same tests. The command list on the following pages provides the minimum reusable verification set.

Acceptance matrix use

Use the matrix as a release checklist. A failed row should identify the responsible layer first rather than trigger broad configuration changes across the topology.

Check	Expected evidence	If not correct
Configuration	Relevant command or endpoint screen matches this section	Return to the owning device/configuration
Behavior	Gateway, routing or service test passes	Trace the path one layer at a time
Record	Final state is saved and documented	Save the known-good state before further changes

Extended engineering note. The value of this section is not limited to the command itself. The project uses each feature as part of an observable service chain. A correct VLAN, for example, matters because it places the endpoint in the correct broadcast domain; a correct route matters because it moves the packet to the next control-plane boundary; a correct NAT rule matters because the return traffic must map back to the private source. The same evidence-first principle applies throughout the design.

For maintenance, compare the current state against the documented baseline before making changes. Capture the relevant verification output, make one controlled change, and repeat the test. This creates a simple audit trail and protects the last known-good state. In this lab, that discipline was especially useful for DHCP Snooping, AP uplink VLAN membership, default routes and NAT source coverage.

37 | Packet Tracer Command Compatibility

Verified syntax used in the final lab

Command family	Validated form	Use
Configuration display	show running-config	Full configuration
Section search	show running-config section ...	Targeted configuration
Filtered search	show running-config include ...	Targeted lines
Interface state	show interfaces status	Switch port overview
Trunk verification	show interfaces trunk	802.1Q/trunk state
VLAN overview	show vlan brief	Access VLAN membership
MAC table	show mac address-table	Dynamic/static MAC location
DHCP Snooping	show ip dhcp snooping	Snooping state
DHCP bindings	show ip dhcp snooping binding	Snooped leases

Several command variants commonly seen on other Cisco IOS references were rejected by the Packet Tracer switch image. Examples included `show running-config interface ...` and VLAN-filtered MAC-table syntax. The final documentation therefore uses only forms actually accepted during the build.

This matters because troubleshooting time is part of operational quality. When an engineer is under pressure, a command that is not supported can create a false branch in the investigation. The normalized syntax prevents the tool itself from becoming the source of ambiguity.

Practical rule

When unsure, start with a broadly supported command such as `show running-config`, `show interfaces status` or `show vlan brief`, then narrow the investigation based on the observed state.

CLI compatibility

The final command set is intentionally conservative. Prefer commands validated on the actual lab image over syntactically valid commands copied from another IOS family.

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Configuration	Relevant command or endpoint screen matches this section	Return to the owning device/configuration
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38 | Configuration Backup and Recovery

Preserving the final working Packet Tracer state

Every major configuration change in the build was followed by `write memory`. The Packet Tracer `.pkt` file is still the primary recovery artifact because it contains the complete topology and simulator state.

```
write memory
Building configuration...
[OK]
Ctrl+S -> save the Packet Tracer project
```

Recommended backup item	Reason
Final `.pkt` file	Contains topology plus simulator configuration/state
This PDF documentation	Records design, implementation and validation logic
Critical screenshots	Provide visual evidence of key configurations and tests
Command archive	Supports rebuild if the project must be recreated

A practical recovery process is simple: reopen the last known-good `.pkt`, inspect the device configuration, run the verification matrix, and compare the current state with the documented addressing and port assignments. Changes should be made only after the baseline is understood.

In a production environment, configuration backups would normally be version-controlled, encrypted and stored off-device. The Packet Tracer lab does not require that infrastructure, but the same discipline applies conceptually: keep a known-good baseline and make changes auditable.

Recovery discipline

Save after major milestones and preserve the last known-good `.pkt` file. A clear baseline makes experiments reversible and keeps troubleshooting bounded.

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39 | Operations Runbook

Day-2 checks, fault isolation and maintenance routine

The final project can be operated using a short repeatable routine. Start with interface health, then validate VLANs and trunks, then routing, then client addressing and finally the Internet/NAT path.

Sequence	Command / action	Purpose
1	show interfaces status	Identify down, err-disabled or wrong-VLAN ports
2	show vlan brief	Confirm access-port membership
3	show interfaces trunk	Confirm allowed/forwarding VLANs
4	show etherchannel summary	Confirm LACP bundle health
5	show ip ospf neighbor	Confirm dynamic routing adjacency
6	show ip route	Check connected, OSPF and default routes
7	show ip dhcp binding	Confirm branch leases
8	show ip nat translations	Confirm active PAT sessions
9	Endpoint gateway ping	Isolate local VLAN/access issues
10	Endpoint ping 8.8.8.8	Confirm end-to-end Internet path

For a wireless issue, add a client-side check of SSID association, adapter state and DHCP address. For an IoT issue, remember that the configuration screen is under Wireless0 rather than the Desktop tab used by PCs.

Escalation rule

Do not modify multiple layers at once. Make one change, verify the immediate dependency, and then continue. This is how the project was brought from multiple simultaneous symptoms to a stable final state.

Operations discipline

Routine checks should start with interface and VLAN health, continue through routing, DHCP and NAT, and end with endpoint tests. This ordering minimizes unnecessary changes.

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40 | Final Acceptance and Project Summary

Complete handover record

The final network provides a complete enterprise lab architecture with HQ and branch segmentation, Layer-3 routing, dynamic OSPF exchange, branch DHCP, Guest isolation, secure management, Layer-2 protection, wireless access, dedicated IoT networking, PAT and a self-contained Internet test endpoint.

Capability	Final verified state
HQ routing	SW-CORE SVIs active; OSPF to R-HQ; explicit default to R-HQ
Branch routing	R-BR router-on-a-stick; OSPF to R-ISP; branch default to R-ISP
Switching	Branch LACP Port-channel4 carrying required VLANs
Security	Guest ACL, SSHv2, Port Security, PortFast/BPDU Guard, DAI
DHCP	Branch leases verified across multiple VLANs
HQ Wi-Fi	10.10.90.21 and .22 validated
Branch Wi-Fi	10.20.90.23 and .22 validated
HQ IoT	10.10.80.22, .23 and .24 validated
NAT/PAT	10.10.255.1 translations observed
Internet simulation	8.8.8.8 reachable from validated client paths

The project also captures realistic troubleshooting lessons. Branch DHCP failures were resolved by understanding DHCP Snooping behavior and the access VLAN. HQ Internet reachability was repaired by completing the default-route chain and expanding NAT source coverage. These incidents make the project stronger as an engineering portfolio artifact because they demonstrate diagnostic reasoning rather than only successful configuration entry.

Final acceptance

The Packet Tracer project can be treated as complete when the `.pkt`` file is saved, the verification matrix passes, the documented wireless and IoT addresses are present, and the final screenshots and command outputs are retained with the project.

End of engineering documentation. This 40-page edition is intentionally content-dense and continuous so that each page is occupied by useful project material rather than placeholder spacing.

Final handover

The project is accepted when the saved topology, documentation and validation evidence agree. Future changes should be documented as deltas from this baseline rather than mixed into the original record.

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